Surface Area and Volume: Mixed Practice

Name _____ Date _____ Calculator allowed Version 1

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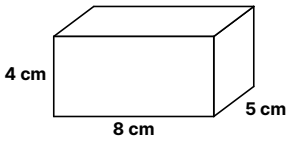
KEY WORDS surface area *(horahanga mata)* · volume *(rōrahi)* · capacity *(rōrahi)* · net *(raupapa mata)*

A Both at once ●●○

Remember: Surface area is the total area of the faces (square units). Volume is the space inside (cubic units). Write both formulas down before you start.

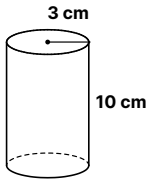
1. A cuboid 8 cm by 5 cm by 4 cm.

a) Find its surface area. **b)** Find its volume.



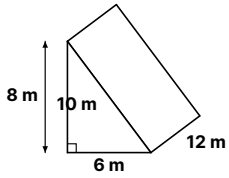
2. A cylinder with radius 3 cm and height 10 cm. Give answers to 1 decimal place.

a) Find its surface area. **b)** Find its volume.



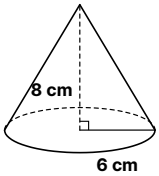
3. A triangular prism. Its end is a right-angled triangle with sides 6 m, 8 m and 10 m, and it is 12 m long.

a) Find its surface area. **b)** Find its volume.



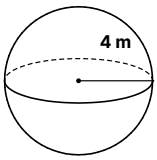
4. A cone with radius 6 cm and height 8 cm. Give answers to 1 decimal place.

a) Find its surface area. **b)** Find its volume.



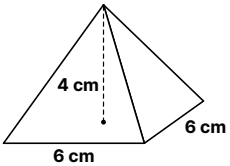
5. A sphere with radius 4 m. Give answers to 1 decimal place.

a) Find its surface area. **b)** Find its volume.



6. A square-based pyramid with base edges of 6 cm and height 4 cm. Give answers to 1 decimal place where needed.

a) Find its surface area. **b)** Find its volume.



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B Which is bigger? ●●●

> **Remember:** Work out the quantity for each solid, then compare. Make sure both answers use the same unit.

7. Cylinder A has radius 4 cm and height 6 cm. Cube B has edges of 5 cm and prism C is 5 cm by 5 cm by 5 cm.

a) Find the volume of A, to 1 decimal place.

b) Find the volume of C.

c) Which holds more, A or C? By how much?

8. A sphere has a radius of 6 cm. A cube has edges of 5 cm.

a) Find the surface area of the sphere, to 1 decimal place.

b) Find the surface area of the cube.

c) Which has the larger surface area?

C Working backwards ●●●

> **Remember:** Decide whether the question is about surface area (square units) or volume (cubic units), write that formula, and solve.

9. Find the missing length.

a) A cube has a volume of 343 cm^3. Find its edge length.

b) A cube has a surface area of 486 cm^2. Find its edge length.

c) A cylinder has a volume of 1,000 cm^3 and a height of 8 cm. Find its radius, to 1 decimal place.

d) A sphere has a surface area of 400 cm^2. Find its radius, to 1 decimal place.

D Problems ●●●

10. A cylindrical rainwater tank, open at the top, has a radius of 0.8 m and a height of 1.5 m.

a) How many litres does it hold, to the nearest litre? (1 m^3 = 1000 L.)

b) The outside (the curved side and the base) is painted. Paint costs $7 per square metre. Find the cost, to the nearest dollar.

11. A grain silo is a cylinder of radius 3 m and height 10 m with a hemisphere of the same radius on top. Find its volume, to the nearest cubic metre.

12. A food tin is a cylinder with radius 4 cm and height 11 cm.

a) Find the area of metal in the tin, to 1 decimal place.

b) Find how many millilitres it holds, to the nearest millilitre. (1 cm^3 = 1 mL.)

★ Extension

13. A solid is enlarged by a scale factor of 3. What happens to its surface area and to its volume? Use a cube with edges of 2 cm to check.

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14. Why do small animals lose heat faster than large ones? Explain using surface area and volume.

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Surface Area and Volume: Mixed Practice

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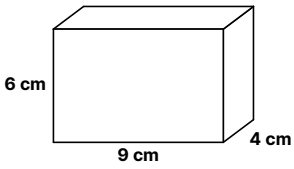
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A Both at once ●●○

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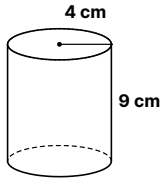
1. A cuboid 9 cm by 4 cm by 6 cm.

a) Find its surface area. **b)** Find its volume.



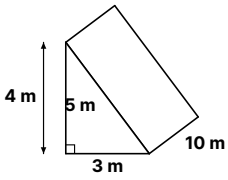
2. A cylinder with radius 4 cm and height 9 cm. Give answers to 1 decimal place.

a) Find its surface area. **b)** Find its volume.



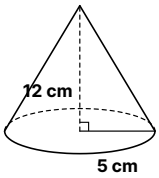
3. A triangular prism. Its end is a right-angled triangle with sides 3 m, 4 m and 5 m, and it is 10 m long.

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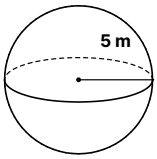
4. A cone with radius 5 cm and height 12 cm. Give answers to 1 decimal place.

a) Find its surface area. **b)** Find its volume.



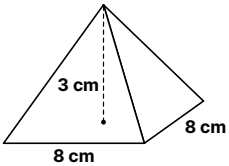
5. A sphere with radius 5 m. Give answers to 1 decimal place.

a) Find its surface area. **b)** Find its volume.



6. A square-based pyramid with base edges of 8 cm and height 3 cm. Give answers to 1 decimal place where needed.

a) Find its surface area. **b)** Find its volume.



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B Which is bigger? ●●●

> **Remember:** Work out the quantity for each solid, then compare. Make sure both answers use the same unit.

7. Cylinder A has radius 3 cm and height 8 cm. Cube B has edges of 4 cm and prism C is 4 cm by 4 cm by 6 cm.

a) Find the volume of A, to 1 decimal place.

b) Find the volume of C.

c) Which holds more, A or C? By how much?

8. A sphere has a radius of 8 cm. A cube has edges of 6 cm.

a) Find the surface area of the sphere, to 1 decimal place.

b) Find the surface area of the cube.

c) Which has the larger surface area?

C Working backwards ●●●

> **Remember:** Decide whether the question is about surface area (square units) or volume (cubic units), write that formula, and solve.

9. Find the missing length.

a) A cube has a volume of 512 cm^3. Find its edge length.

b) A cube has a surface area of 384 cm^2. Find its edge length.

c) A cylinder has a volume of 1,500 cm^3 and a height of 10 cm. Find its radius, to 1 decimal place.

d) A sphere has a surface area of 600 cm^2. Find its radius, to 1 decimal place.

D Problems ●●●

10. A cylindrical rainwater tank, open at the top, has a radius of 1 m and a height of 1.2 m.

a) How many litres does it hold, to the nearest litre? (1 m^3 = 1000 L.)

b) The outside (the curved side and the base) is painted. Paint costs $8 per square metre. Find the cost, to the nearest dollar.

11. A grain silo is a cylinder of radius 2.5 m and height 12 m with a hemisphere of the same radius on top. Find its volume, to the nearest cubic metre.

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a) Find the area of metal in the tin, to 1 decimal place.

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★ Extension

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